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## 6. Optimizing the design

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Part B due Mar 6, 2024 20:30 IST

The next step is to study the system's behavior at other forcing frequencies, as we want to minimize the largest amplitude of the building's oscillations that can occur in the range  $\Omega \in [0.7, 1.3]$ .

LTI Consumer (External resource) (1.0 points possible)

Sweeping through input frequencies

The script below is designed to compute and plot the swaying amplitude of the building over a range of forcing frequencies. The amplitude of the building without a TMD will also be plotted for comparison. To complete the script, you need to:

- 1. Define the variable `omSweep` to be a row vector containing 50 equally spaced points over the range  $\Omega \in [0.7, 1.3]$  using the `linspace` function.)
- 2. Copy the relevant lines of code from the previous problem to solve the system using ODE45.

To test the script, use the parameters:

$$\begin{aligned} m_1 &= 1, & m_2 &= 0.05, \\ k_1 &= 1, & k_2 &= 0.02, \\ b_1 &= 0.001, & b_2 &= 0.02. \end{aligned}$$

Use the initial conditions  $\mathbf{x}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ ,

and solve the system for times  $t$  on the interval  $[0, 7000]$  using `linspace` with 35000 points. If your script changes the swaying amplitude of the building for each forcing frequency  $\Omega$  in the range  $[0.7, 1.3]$ .

Script ?

 Save  Reset  MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 load('Reference.mat')
2
3 %Define parameters
4 m1 = 1;
5 m2 = 0.05;
6 k1 = 1;
7 k2 = 0.02;
8 b1 = 0.001;
9 b2 = 0.02;
10 omSweep = linspace(0.7, 1.3, 50);
11
12 %Numerically solve DE at each forcing frequency
13 for i=1:length(omSweep)
14
15     om = omSweep(i);
16
17     %Copy code to solve DE here
18     %Numerically solve DE
19     x0 = [0;0;0;0];
20     tvec = linspace(0,7000,35000);
21     A = [0,0,1,0;0,0,0,1;-(k1+k2)/m1,k2/m1,-(b1+b2)/m1,b2/m1;k2/m2,-k2/m2,b2,0];
22     [t,x] = ode45(@(t,x) A*x + [0;0;sin(om*t)/m1;0],tvec,x0);
23
24     %Compute steady state solution and extract building oscillation amplitude
25     lt = length(t);
26     per = 2*pi/om;
27     [~,idx] = min(abs(t-(t(end)-5*per)));
28     ampBuilding(i) = max(x(idx:lt,1));
29
30 end
31
32 %Plot
33 plot(OmRef,ampRef,'r'); hold on;
34 plot(omSweep,ampBuilding,'k');
35 xlim([0.7,1.3]); ylabel('Max. Amp. of Building'); xlabel('\Omega');
36 legend('Without TMD','With TMD')
37
```

 Run Script 

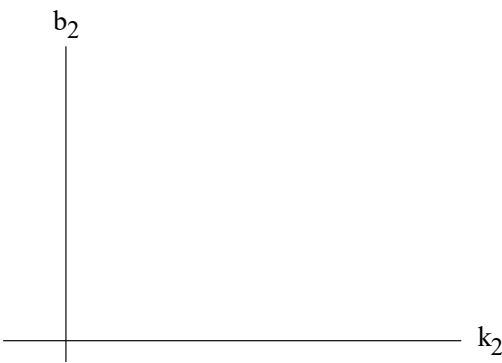
 Calculator

✔ Correct definition of OmSweep

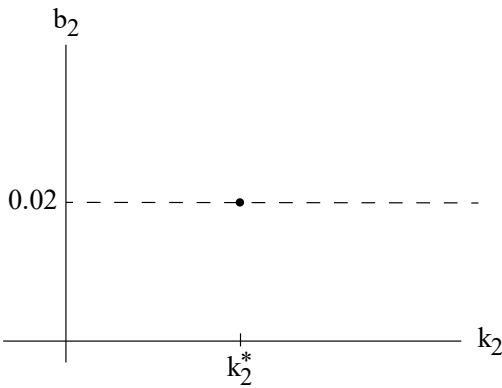
### Optimization problem

The optimal values of  $k_2$  and  $b_2$  will minimize the maximum value of the black response curve.

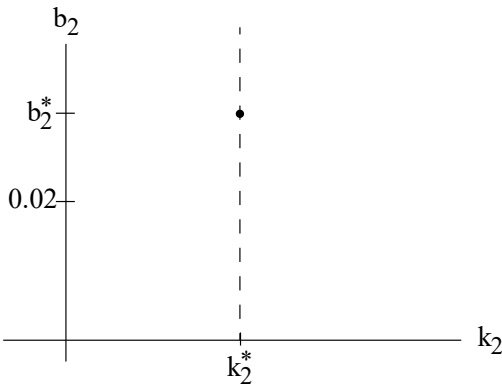
This is a two-dimensional minimization problem. We have a plane of possible values for  $k_2$  and  $b_2$  that we must attempt to minimize this problem over.



However, in our example where the building has a small damping constant,  $b_1$ , it can be shown that the parameters  $k_2$  and  $b_2$  can be optimized independently of each other. Therefore, our approach is to fix  $b_2 = 0.02$  and first determine the optimal value,  $k_2^*$ , of the spring constant.



Next we fix  $k_2 = k_2^*$  at this optimal value and determine the optimal value,  $b_2^*$ . (You can test that if you then fix  $b_2^*$ , and try to re-optimize  $k_2$ , your answer does not change.)



### MATLAB exploration 1

1 point possible (graded)  
Copy the script you used in the previous problem (*Sweeping through input frequencies*) to MATLAB online or on your computer. You will also need to save and copy the file 'Reference.mat'. You can do this by typing

```
url = 'https://courses.edx.org/asset-v1:MITx+18.033x+1T2018+type@asset+block@Reference.mat';  
websave('Reference.mat', url);
```

into your MATLAB command window.

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While keeping all of the other parameters constant, change  $k_2$  and see what effect it has on the response curve of the building. Based on your observations, enter the optimal value,  $k_2^*$ , that is accurate to three decimal places.

$k_2^* =$

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An error occurred. Unable to load drag and drop problem.

## MATLAB exploration 2

1 point possible (graded)  
Now that you have found the optimal spring constant,  $k_2^*$ , the last step is to determine the optimal damping constant,  $b_2^*$ . Using your value of  $k_2^*$  and keeping all of the other parameters constant, vary  $b_2$  and see what effect it has on the response curve of the building. Like in the previous problem, based on your observations, enter the optimal value,  $b_2^*$ , that is accurate to three decimal places.

$b_2^* =$

Submit

You have used 0 of 10 attempts

An error occurred. Unable to load drag and drop problem.

We have achieved our goal and have determined the optimal spring and damping constants that minimize the swaying amplitude of the building over a range of forcing frequencies! You should double check that using the value of  $b_2$  found in the previous step and then optimizing  $k_2$  will give you the same result. (A general numerical approach to optimizing two parameters is to iteratively optimize each parameter independently, and using the new value to re-optimize the other parameter until no changes are detected.)

Note that this problem is a toy model. In reality, there are often many other factors that engineers must consider when designing a TMD for a building. For example:

- The building might sway in multiple dimensions and might also twist around its vertical axis.
- The building might need to withstand forcing from the ground due to seismic activity, which can produce a very different response compared to wind forcing.

## 6. Optimizing the design

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